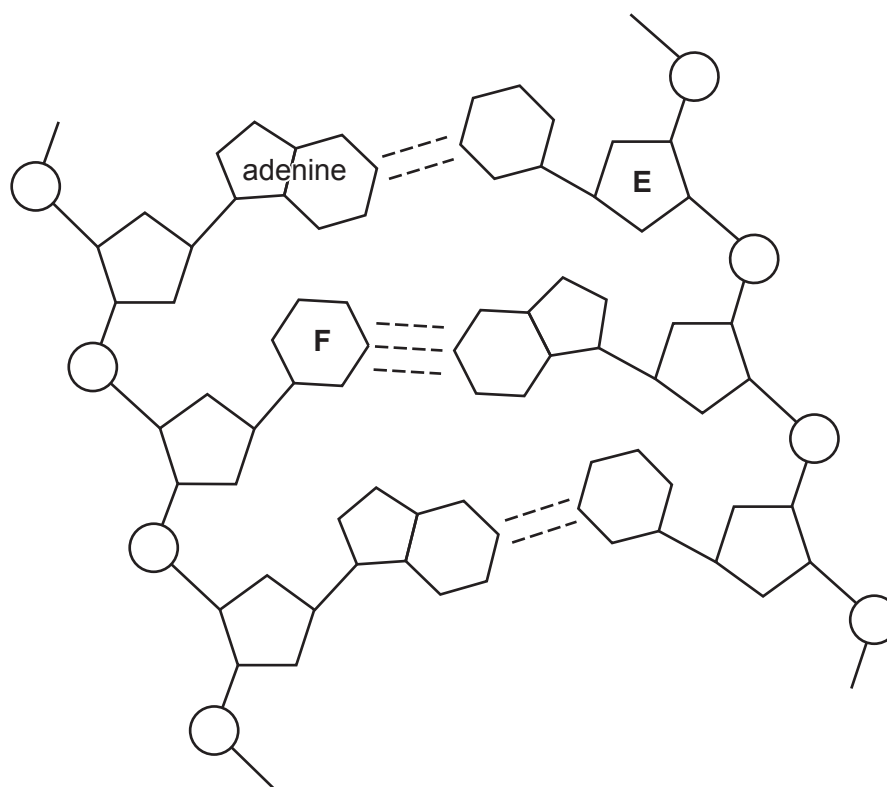


6 Fig. 6.1 is a diagram showing the structure of part of a DNA molecule.



key

----- one hydrogen bond

Fig. 6.1

(a) (i) Identify structure **E** and structure **F** in Fig. 6.1.

E

F

[2]

(ii) On Fig. 6.1 draw a circle around **one** nucleotide.

[1]

(iii) State the name of the covalent bond that links two nucleotides together.

..... [1]

- (b) Fig. 6.2 shows the RNA base sequence of a short length of primary transcript.

Complete Fig. 6.2 by writing the DNA base sequence of the template strand used to form the primary transcript.

DNA base sequence used to form the primary transcript				
primary transcript	G G U	G C U	A A U	C U A

Fig. 6.2

[1]

- (c) In eukaryotic cells, the primary transcript is modified to form mRNA.

Explain how the primary transcript is modified to form mRNA.

.....

.....

.....

.....

..... [2]

- (d) The mRNA strand is translated at the ribosome to form a polypeptide.

Describe how the process of translation results in the formation of a polypeptide.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [5]

[Total: 12]